

BRAUNSCHWEIG, Germany

WMO: 103480

Lat: 52.2914N

Lon: 10.4464E

Elev: 82

StdP: 100.34

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.4	-9.7	-14.5	1.1	-11.1	-12.1	1.3	-8.7	11.2	6.7	10.2	5.2	2.2	90	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	28.7	19.0	26.7	18.1	24.9	17.3	19.9	26.1	18.9	25.0	18.1	23.4	3.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	22.1	16.7	12.0	21.3	15.8	11.3	20.5	57.4	26.2	54.0	25.0	51.2	23.5	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.4	DB	-14.0	32.5	5.8	2.4	-18.2	34.2	-21.6	35.6	-24.9	36.9	-29.1	38.6	(4)
(5)			WB	-14.3	22.1	5.8	0.9	-18.4	22.8	-21.8	23.3	-25.0	23.8	-29.2	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	0.9	0.5	4.3	8.1	12.7	15.7	17.7	17.8	13.7	9.8	5.4	2.7
	DBStd	7.22	5.63	5.22	4.10	3.72	3.58	3.36	3.04	2.91	2.44	3.64	3.63	3.98
	HDD10.0	1247	283	268	181	79	14	1	0	0	2	49	141	229
	HDD18.3	3431	541	500	436	307	178	94	50	46	140	266	387	486
	CDD10.0	934	0	1	3	22	98	171	238	242	113	42	4	1
	CDD18.3	77	0	0	0	0	3	13	29	29	1	1	0	0
	CDH23.3	744	0	0	0	1	39	136	282	268	15	2	0	0
	CDH26.7	164	0	0	0	0	2	29	62	72	1	0	0	0
Wind	WSAvg	3.5	4.4	3.9	3.9	3.4	3.2	3.1	3.0	2.9	3.3	3.4	3.7	3.8
Precipitation	PrecAvg	671	58	44	45	42	58	57	75	71	56	54	53	57
	PrecMax	931	108	90	111	84	189	94	194	176	142	147	109	129
	PrecMin	514	9	12	10	8	18	19	26	19	9	15	1	19
	PrecStd	103	29	22	27	18	38	22	37	33	33	29	28	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	14.6	17.4	21.8	26.6	30.2	30.1	31.1	26.0	22.7	14.7	12.7
		MCWB	9.2	10.6	10.7	13.4	16.2	19.0	19.5	19.9	17.5	16.7	10.8	9.7
	2%	DB	10.1	11.6	13.6	19.1	24.3	26.8	28.5	28.6	22.6	19.1	13.0	11.5
		MCWB	8.2	8.3	9.4	12.1	15.3	18.4	19.3	18.6	16.4	14.3	10.4	9.1
	5%	DB	8.8	8.7	11.6	16.9	22.0	24.6	26.6	26.2	20.2	16.6	11.7	10.0
		MCWB	6.9	6.3	8.3	10.7	14.6	17.1	18.5	17.9	15.2	13.3	9.7	8.2
	10%	DB	7.3	6.9	10.0	14.7	19.8	22.4	24.6	24.1	18.5	15.0	10.6	8.3
		MCWB	5.6	5.0	7.5	9.7	13.8	16.0	17.8	17.2	14.2	12.3	8.9	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	10.8	11.0	14.0	17.6	20.6	21.5	20.7	18.6	17.1	11.8	10.2
		MCDB	11.3	14.2	16.0	19.7	23.8	26.9	28.2	28.8	23.1	21.2	13.6	11.8
	2%	WB	8.4	8.6	9.9	12.5	16.2	19.0	20.2	19.4	16.9	15.2	10.8	9.4
		MCDB	10.0	11.2	12.9	17.8	22.4	25.5	26.3	25.7	20.7	18.4	12.3	11.0
	5%	WB	6.9	6.5	8.8	11.3	15.2	17.8	19.2	18.7	15.9	13.7	9.9	8.4
		MCDB	8.6	8.4	11.0	16.2	21.1	23.1	25.2	24.4	19.7	15.8	11.5	10.0
	10%	WB	5.6	5.1	7.7	10.1	14.0	16.6	18.1	17.8	14.8	12.7	8.9	6.9
		MCDB	7.2	6.8	9.9	14.2	18.9	21.3	23.5	22.7	17.9	14.7	10.6	8.2
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	6.3	8.5	9.4	8.9	9.5	9.5	7.8	6.6	5.1	4.0
		MCDBR	5.3	7.5	9.3	12.4	13.1	13.1	13.4	13.6	11.6	9.2	7.1	5.7
		MCWBR	4.2	5.3	5.7	6.8	6.6	6.2	6.2	6.0	6.4	5.7	5.1	4.6
	5% WB	MCDBR	5.1	6.9	8.2	10.9	11.7	11.4	11.9	11.1	9.8	8.4	6.5	5.5
		MCWBR	4.4	5.2	5.3	6.3	6.4	6.0	6.3	5.5	5.9	5.5	5.0	4.7
Clear-Sky Solar Irradiance	taub	0.317	0.339	0.375	0.411	0.414	0.411	0.428	0.420	0.385	0.358	0.337	0.312	
	taud	2.364	2.347	2.264	2.212	2.242	2.270	2.232	2.268	2.357	2.417	2.399	2.362	
	Ebn at Noon	677	769	807	821	838	843	819	804	791	739	646	613	
	Edh at Noon	61	82	107	127	130	127	130	119	97	76	59	52	
All-Sky Solar Radiation	RadAvg	0.67	1.37	2.44	3.99	4.91	5.29	4.97	4.22	3.09	1.76	0.79	0.51	
	RadStd	0.07	0.19	0.25	0.47	0.56	0.42	0.68	0.36	0.36	0.18	0.09	0.04	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (48 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page